

Tareq Shawabkeh

AI, ML, Data, CV, Software and Backend Integration Engineer

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SUMMARY

Computer Engineering Fresh graduate specializing in **Artificial Intelligence, ML**, Data and **Backend Software Engineering**. with strong hands-on experience in building real-time, AI-integrated systems and API-driven applications. Experienced in developing production-style systems from design to deployment, including scalable **backend** architectures (Python / Fast API), secure data validation pipelines, and enterprise-grade REST APIs. Proven ability to deliver award-winning solutions, highlighted by engineering an **AI-automated trading platform (Graduation Project 4/4) that won First Place at the NTP2025 National Competition**. Strong in machine learning workflows, system design, and transforming data into reliable, production-oriented solutions. Seeking an Entry level or Internship opportunities to grow in AI, data engineering or Backend Software Engineer within a practical, real-world environment. with practical experience gained through internships and academic teaching assistance.

EDUCATION

B.Sc. in Computer Engineering University of Jordan — Amman, Jordan | **Graduated: January 2026**

Graduation Project: “Real-Time and AI-Based Automated Trading App” (ML Bot + Backend API + Flutter App).

Relevant Coursework: Python, Machine Learning, Data Structures, Network Programming, Java, Probability & Statistics.

SKILLS

Programming Languages: *Python*, Java, C++, SQL, Dart-Flutter

Backend & Systems: FastAPI, REST APIs, WebSocket’s, Multithreading, TCP/IP, Async Processing, Client–Server Architecture, Secure API Design & Data Validation.

Databases: *PostgreSQL*, *SQLite*, SQLAlchemy, JPA/Hibernate.

AI / Data: Scikit-learn, TensorFlow, data processing with Pandas, NumPy, Data preprocessing & feature engineering.

Tools & Practices: Docker, Postman, Swagger/OpenAPI, Git, VS Code, NetBeans, API Design, Real-Time Trading systems, Secure Key Management, Client–Server Architecture.

ACHIEVEMENTS

- 🏆 First Place Winner — NTP2025 Competition (Jordan)
- Top 10 Graduation Projects in School of Engineering & IT — University of Jordan

PROJECTS

1. **Real-Time & AI-Based Automated Trading App (🏆 1st Place Winner NTP2025)**
Graduation Project | *University of Jordan* | *Oct 2024 – May 2025*

Designed and implemented a full-stack, real-time automated trading system for cryptocurrency markets, combining machine learning models, trading strategy, with a high-performance backend - ensuring multi-user support, encrypted API key handling and live execution infrastructure (tracks performance in real time).

The system processes live market data, executes strategies and **custom ML model** were built from scratch that achieved **162% Total Return** Vs. only **8.44% Max Drawdown** in long back-testing after fine tuning and optimization on the best model we got.

Key Contributions:

- Architected a modular system integrating a Flutter mobile app, **Python - FastAPI Backend**, SQLite database, and broker APIs for end-to-end automation
- Developed a real-time trading engine that continuously processes live market data, generates signals, and executes trades automatically
- Designed an LSTM-based machine learning model for time-series prediction, evolving from price regression to feature-based classification using indicators (MACD, RSI)
- Designed and implemented the backend using FastAPI with async processing and thread-based trading loops.
- Built secure multi-user architecture with encrypted API key storage and session persistence.
- Integrated multiple trading strategies (SMA, MACD, RSI, ADX + ML - based model) with real-time execution.
- Implemented WebSocket-based live updates for trades, balance, and performance tracking.
- Collaborated on Flutter mobile app for strategy control and monitoring.

Tech Stack: FastAPI, SQLite, SQLAlchemy (Async), WebSocket, Threading, TensorFlow, Cryptography, Flutter.

2. **Car Price Prediction using Advanced Regression Techniques (AI Course)** *Jan 2024 – Mar 2024*

- Built an end-to-end car price prediction pipeline starting from raw CSV data to model evaluation.
- Designed a CustomTransformer preprocessing pipeline to handle mixed features: numerical attributes were scaled, while categorical data (make, model, color, state) was encoded in a structured and reusable way.
- Used the same preprocessing pipeline across training, validation, and testing to ensure consistent data flow and prevent data leakage.
- Trained and compared multiple models, then extended the work to neural networks and deep learning architectures, including CNN-based feature learning.
- Evaluated model performance using RMSE and validation loss to select stable and generalizable configurations.

3. **FX Deals Importer System (Spring Boot, PostgreSQL)**

- Built REST APIs for real-time FX deal submission and bulk CSV import.
- Designed layered architecture (Controller, Service, Repository, DTO, Entity) to ensure clean separation of concerns.
- Documented APIs using Swagger and tested endpoints using Postman.
- Dockerized the application for consistent deployment.

4. Client–Server Medical Consultation Platform (Java Backend, TCP/IP, Multithreading)

Course Project – Network Programming, Spring 2024

Developed a multithreaded Java Backend Application simulating a real-time **medical (Patient – Doctor) consultation system**. The server handled two user roles-doctors (with secure login, registration, and specialty-based ticket queues) and patients (submitting general or directed medical inquiries).

- Implemented TCP/IP-based client–server communication.
- Designed concurrent server handling for multiple doctors and patients.
- Built secure login, ticketing, and queue management system.
- Doctor registration/login and real-time ticket queue management (e.g., 5 patients per doctor)
- Patient ability to browse online doctors by specialty and submit general or direct tickets
- Implemented FIFO scheduling, timeout handling, and availability checks.

5. Aerial Image Object Detection Pipeline – Cars Detection (Computer Vision)

2026 April

- **Developed** an end-to-end computer vision pipeline using YOLOv8 Nano to accurately detect and track vehicles in high-resolution, top-down aerial imagery.
- **Engineered** a robust data ingestion module that dynamically parses PASCAL VOC XML annotations to YOLO format while enforcing deterministic dataset splitting.
- **Achieved** an outstanding model performance of 99.0% mAP@50, specifically optimizing the architecture for lightweight, edge-computing environments like drones.
- **Designed** a flexible Command-Line Interface (CLI) using argparse to facilitate seamless, path-independent inference on both test sets and unseen "wild" data.
- **Ensured** strict cross-platform reproducibility by building a pinned, CPU-optimized Python environment, completely eliminating dependency and architecture conflicts.

EXPERIENCE

Software Engineering Intern - Link Business Solutions Company.

Amman, Khalda, Jordan | Summer 2025

- Contributed to web system implementation using HTML, CSS, JavaScript, and Java within a professional development environment.
- Participated in transforming UI designs into functional web pages and interactive while following company standards and clean coding practices (e.g., TO-DO-List).
- *Worked with backend developers to integrate frontend components with server-side logic (e.g., **The FX deals Backend Project**).*
- Gained exposure to real production workflows, requirement changes, and team-based delivery.

Teaching Assistant - Python Programming Lab

University of Jordan | 2025 – 2026

- Assisted undergraduate students in Python labs by explaining programming concepts, debugging code, and guiding problem-solving approaches.
- Supported students during lab sessions and assignments, helping them understand logic, syntax, and program structure.
- Developed **strong communication** and **teaching skills** by simplifying complex programming ideas for different learning levels.
- Provided feedback on student code and corrected logical and runtime errors.